

SIC AI Chapter Quiz: 07

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Question 01

- **Answer:** Option 4 ("The same method can be applied to various languages.")
- **Reasoning:** Text mining techniques rely heavily on linguistic structures (grammar, vocabulary, syntax) which vary significantly between languages. A method or library designed for English (e.g., standard NLTK tokenizers) will often fail or require significant modification to work on languages like Korean, Arabic, or Chinese.

Question 02

- **Answer:** Option 4 ("It is possible to know if customers will leave next time.")
- **Reasoning:**
 - **Options 1, 2, and 3** describe the core functions of Social Network Analysis (SNA): finding communities (groups), identifying influencers (centrality), and tracking structural changes over time.
 - **Option 4** describes **Churn Prediction**, which is a predictive modeling (classification) task. While SNA can provide data *for* such a model, the analysis of the network structure itself does not directly predict future individual behavior like churning.

Question 03

- **Answer: Stemming**
- **Definition:** A text normalization technique in Natural Language Processing (NLP) that reduces words to their base or root form (the stem) by removing suffixes and prefixes (e.g., converting "running", "runs", and "ran" to "run").

Question 04

- **Answer: Corpus** (or **Corpora**)
- **Definition:** A large, structured collection of texts (written or spoken) used as a dataset for linguistic analysis and statistical processing in Natural Language Processing (NLP).

Question 05

```
print("Explanation:")
print("- TF measures how often a specific term (word) appears in a document.")
print("- It assumes that words appearing frequently are important to that document.\n")

print("Formula:")
print("TF(t, d) = (Number of times term 't' appears in document 'd') / (Total number of terms in document 'd')")
```

Explanation:

- TF measures how often a specific term (word) appears in a document.
- It assumes that words appearing frequently are important to that document.

Formula:

$$TF(t, d) = (\text{Number of times term 't' appears in document 'd'}) / (\text{Total number of terms in document 'd'})$$

Question 06

```
import numpy as np
import nltk
from nltk.corpus import stopwords
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics import pairwise_distances
```

```

# Ensure stopwords are available
nltk.download('stopwords')

# 1. The Corpus (Documents)
doc = [
    "AI is a rapidly advancing field that involves the development of intelligent machines.", # 0
    "Machine learning is a subset of AI that focuses on training machines to learn from data.", # 1
    "Deep learning is a subfield of machine learning that utilizes neural networks with multiple layers.", # 2
    "AI applications are found in various industries, including healthcare, transportation, and entertainment.", # 3
    "Ethical consideration plays an important role in AI development.", # 4
    "AI has the potential to revolutionize many aspects of society.", # 5
    "AI systems such as chatbots and virtual assistants are becoming more common.", # 6
    "Natural language processing is a branch of AI that allows machines to understand human language.", # 7
    "The field of computer vision aims to enable machines to understand visual information.", # 8
]

# 2. Pre-processing
doc = [x.lower() for x in doc]

# 3. TF-IDF Vectorization
# 'AI' and 'machines' are likely removed because they appear in > 3 docs.
vectorizer = TfidfVectorizer(max_features=18, min_df=1, max_df=3, stop_words=stopwords.words('english'))
X = vectorizer.fit_transform(doc).toarray()

# 4. Calculate Cosine Similarity
# The provided code imports pairwise_distances.
# Cosine Distance = 1 - Cosine Similarity.
# We want the SMALLEST distance (which equals HIGHEST similarity).
dist_matrix = pairwise_distances(X, metric='cosine')

# Set diagonal to infinity so we don't pick a document as closest to itself
np.fill_diagonal(dist_matrix, np.inf)

# Find the indices of the minimum distance
min_dist_idx = np.unravel_index(np.argmin(dist_matrix), dist_matrix.shape)
idx1, idx2 = min_dist_idx

print(f"--- Most Similar Pair (Indices: {idx1} and {idx2}) ---")
print(f"Statement 1: {doc[idx1]}")
print(f"Statement 2: {doc[idx2]}")
print(f"Cosine Similarity Score: {1 - dist_matrix[idx1, idx2]:.4f}")

--- Most Similar Pair (Indices: 1 and 2) ---
Statement 1: machine learning is a subset of ai that focuses on training machines to learn from data.
Statement 2: deep learning is a subfield of machine learning that utilizes neural networks with multiple layers.
Cosine Similarity Score: 0.8384
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!

```